

PAPER_881: Kozima Expansion Neutron Drop Coupling

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Abstract

Kozima neutron drop coupling in the buoyancy expansion regime. $F_{\text{coupled}} = F_{\text{Kozima}}(\omega_{\text{SCm}}) \times E^+(t) \times \Phi(\omega)$ where the Gaussian cross-section $\sigma_n(\omega) = \sigma_0 \cdot \exp[-(\omega - \omega_{\text{SCm}})^2 / (2\Gamma^2)]$ peaks at the 1.25 THz SCm resonance frequency, coupling phonon-driven LENR neutron drop reactions to the expansion energy engine.

1. Core Equations

$$\begin{aligned}\sigma_n(\omega) &= \sigma_0 \cdot \exp[-(\omega - \omega_{\text{SCm}})^2 / (2\Gamma^2)] \\ F_{\text{Kozima}} &= n \cdot \sigma_n \cdot v_{\text{th}} \cdot \hbar \omega \\ F_{\text{coupled}} &= F_{\text{Kozima}} \times E^+(t)\end{aligned}$$

2. Parameters

Parameter	Default	Description
omega	$2\pi \times 1.25\text{e}12$ rad/s	Angular frequency
T	300 K	Temperature
n_density	$1\text{e}28$ m ⁻³	Number density
E_0	1.0 J	Initial energy scale
t	0.0 s	Time parameter

3. UQFF Integration

This calculator integrates `positive_et_expansion.py` from Session 205 into the CP4 pipeline as class #465.

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